



Integrating in vitro aerosol characterization and cross-species PBPK modeling to predict human lung exposure of inhaled therapeutic proteins

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ABSTRACT

Inhalation therapy provides a rapid onset and avoids first-pass metabolism, making it a key non-invasive delivery strategy for therapeutic proteins. However, predicting pulmonary exposure is challenging due to aerosol properties and lung anatomy. This study is the first to construct an integrated in vitro aerosol characterization and cross-species extrapolation pulmonary physiologically-based pharmacokinetic (PBPK) model for inhaled therapeutic proteins. Using recombinant human interferon- α 1b (rhIFN- α 1b) as a model drug, aerosol aerodynamic properties and delivery performance were evaluated at different concentrations (15–60 μ g/mL) and inhalation devices (mouthpiece vs. facemask). The multiple path particle dosimetry (MPPD) model was used to predict particle deposition in the lungs. Further, a pulmonary PBPK model for rhIFN- α 1b in mice and healthy adults was developed and validated through pharmacokinetic studies. Results showed that delivery efficiency decreased with concentration, and the mouthpiece outperformed the mask. MPPD simulations indicated that the mouthpiece reduced oropharyngeal retention and enhanced alveolar deposition. Larger particle sizes led to reduced alveolar deposition. The PBPK model accurately predicted rhIFN- α 1b exposure in the lungs and serum of healthy adults, with 85% of C_{max} and AUC_{0-inf} values falling within a 0.5–2.0-fold error range. This study provides a quantitative platform for clinical translation and dosing regimen evaluation of inhaled therapeutic proteins.

1. Introduction

Therapeutic proteins have rapidly developed in recent years due to their high efficacy and specificity, with over 240 products approved by the FDA (Kunde et al., 2022; Liang et al., 2020). However, these drugs have extremely low oral bioavailability (<2%) (Nicze et al., 2024), limiting their clinical use to primarily injectable formulations. Unfortunately, injection-based therapies often result in poor patient adherence, particularly in the management of chronic diseases (Chung et al., 2012). In contrast, inhalation therapy, as a non-invasive delivery strategy, offers the advantage of achieving high local concentrations directly in the lungs and potential systemic absorption via the permeable

alveolar epithelium. Currently, several inhaled therapeutic proteins are in clinical development, targeting both respiratory and systemic diseases. For example, Kamada Ltd.'s inhaled Kamada-AAT formulation for the treatment of pulmonary diseases associated with congenital α -1 antitrypsin deficiency has entered Phase III clinical trials (Kunde et al., 2022; Brantly et al., 2025).

The development of inhaled therapeutic protein formulations faces significant challenges due to formulation properties and the complexity of the pulmonary environment. Pulmonary exposure of inhaled therapeutic proteins is influenced by two critical phenomena, deposition and clearance. First, the deposition behavior of inhaled protein drugs in the airways is affected by multiple factors, including aerosol aerodynamic

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particle size, delivery device performance, breathing patterns, and airway anatomy. Deposition site and dose are key determinants of drug exposure (Darquenne, 2012; He et al., 2022; Newman, 2000). Additionally, after inhalation, therapeutic proteins may undergo processes such as mucociliary clearance, nonspecific uptake, and target-mediated drug disposition (TMDD), leading to potential nonlinear pharmacokinetic behaviors. These factors make it difficult to accurately predict the quantitative relationship between the inhaled dose and lung target exposure, thereby limiting the optimization of initial doses and dosing regimens in clinical trials (Liang et al., 2020; Foss et al., 2024; Mager and Jusko, 2001).

To quantitatively assess pulmonary deposition of inhaled formulations, a combined approach of in vitro aerosol characterization and computational modeling is commonly used. Cascade impactors, laser diffraction techniques, and breathing simulators have been routinely used to determine key parameters including aerodynamic particle size, geometric particle size, and aerosol delivery performance (Fisher et al., 2017; Mitchell and Nagel, 2003; Vecellio None et al., 2001). The in vitro techniques have mainly been applied to small molecules or local inhalation formulations. There is still a lack of systematic aerosol characterization under real nebulization conditions for large-molecule therapeutic proteins.

In vitro aerosol parameters alone cannot directly predict the effective drug concentration in lung target areas. Physiologically based pharmacokinetic (PBPK) modeling serves as a crucial tool for translating in vitro aerosol characteristics into in vivo exposure (Shah and Betts, 2012; Thakur et al., 2020; Wong and Chow, 2017). However, there is a clear gap in the development of pulmonary PBPK models for inhaled therapeutic proteins. Although Dhaval et al. developed a PBPK model for therapeutic proteins in adults, the model was validated based on limited intravenous data and did not reflect the actual inhalation delivery process, lacking pharmacokinetic data for the inhalation route and failing to capture deposition distribution and local disposition mechanisms in the lungs (Jagdale et al., 2022).

In this study, recombinant human interferon- $\alpha 1b$ (rhIFN- $\alpha 1b$) was used as a model drug to construct a comprehensive quantitative framework of “in vitro aerosol characterization – pulmonary deposition – cross-species PBPK prediction.” rhIFN- $\alpha 1b$ is mechanistically well-characterized and serves as an archetypal molecule for studying the disposition of therapeutic proteins, with its pulmonary fate

encompassing representative pathways such as receptor-mediated endocytosis and alveolar macrophage-mediated clearance. Additionally, the clinical relevance of inhaled interferons has been well documented (Djukanović et al., 2014; Smaldone, 2018), and comprehensive high-quality pulmonary pharmacokinetic data are available for rhIFN- $\alpha 1b$ from both literature and independent experiments, enabling robust model validation and interspecies scaling. By systematically evaluating delivery characteristics under real nebulization conditions and calibrating and validating the model with pharmacokinetic data from mice and adults, this study successfully achieved the quantitative translation of inhaled protein properties into in vivo exposure. This model provides important quantitative guidance for the clinical translation, initial dose selection, and dosing regimen optimization of inhaled therapeutic proteins, helping to reduce development risks and accelerate their progress.

2. Methods

2.1. In vitro aerosol performance characterization of rhIFN- $\alpha 1b$

The overall process of this study is outlined in Fig. 1. To systematically evaluate the aerosolization of rhIFN- $\alpha 1b$ for injection (Batch number: 202411025, Sinovac Biopharmaceuticals) and provide key input parameters for subsequent lung deposition simulations, the study comprehensively examined the effects of drug concentrations (15, 30, 45, 60 $\mu\text{g/mL}$) and inhalation devices (mouthpiece vs. facemask) on the in vitro aerosol characteristics. All experiments were performed using a jet nebulizer (PARI TurboBOY, PARI GmbH, Germany) for aerosol generation. The aerosol characterization experiments were divided into three parts. First, the geometric particle size distribution was measured using a laser diffraction particle size analyzer (HELOS/INHALER, Sympatec GmbH, Germany), recording the median particle size (D50) and the cumulative volume percentage in the 1–5 μm range (dQ3, 1–5 μm). Second, aerodynamic properties were assessed using a Next-Generation Impactor (NGI, HRH-ZJQ160, Beijing Huironghe, China), from which the mass median aerodynamic diameter (MMAD), geometric standard deviation (GSD), and fine particle fraction (FPF) were calculated. Finally, a breathing simulator (HRH-BRM2100, Beijing Huironghe, China) was used to simulate real delivery conditions under adult breathing patterns, quantifying delivery rate, total delivery, delivery efficiency, and nebulizer residuals under different dosing conditions.

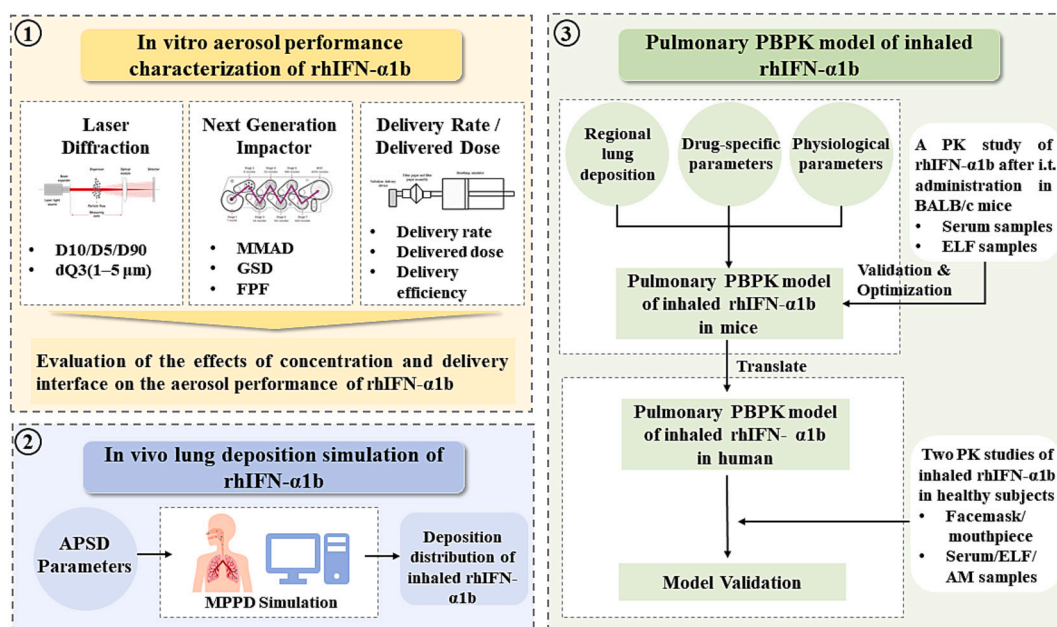


Fig. 1. Flowchart of this study.

The rhIFN- α 1b content in all samples was quantified using a validated double-antibody sandwich enzyme-linked immunosorbent assay (ELISA), with a linear range of 10.0–640 pg/mL. The accuracy (RE%) and precision (CV%) of the method met the acceptance criteria for in vitro analytical method validation. All experiments were independently repeated three times, and data were expressed as mean $\pm$ standard deviation. Statistical analysis was performed using one-way or two-way analysis of variance (ANOVA) with GraphPad Prism software (Version: 10.1.2), where a $P < 0.05$ was considered statistically significant. Detailed experimental procedures and quantitative detection methods are provided in the supplementary methods 1.1–1.2.

2.2. MPPD-based prediction of pulmonary deposition of inhaled rhIFN- α 1b in human

The Multiple Path Particle Dosimetry (MPPD, version 3.04) model was used to simulate the regional deposition of inhaled rhIFN- α 1b in the lungs of adults. The aerodynamic particle size parameters employed in the model were derived from the NGI experiments described in Section 2.1, utilizing MMAD and GSD distributions for all experimental conditions. An adult Weibel airway model was chosen, with an upright posture. The following adult respiratory parameters were set: functional residual capacity (FRC) of 3300 mL, upper airway volume of 50 mL, tidal volume of 500 mL, respiratory rate of 15 breaths/min, and an inspiratory-expiratory ratio of 1:1. For the mouthpiece inhaler, oral inhalation was selected as the delivery method, while for the facemask inhaler, a mouth-nose mixed breathing approach was used. After inputting all the parameters and completing the model calculation, the deposition fractions of rhIFN- α 1b in the 23 airway generations were obtained. The airways were categorized anatomically and functionally as follows: generations 0–9 were considered the upper airways, generations 10–16 were the lower airways, and generations 17–23 were the

alveolar regions. The deposition fractions of rhIFN- α 1b in the oropharynx, upper airways, lower airways, and alveoli were calculated under different particle sizes and inhaler conditions. These results aim to further assess the impact of aerodynamic particle size and inhaler type on deposition patterns at target sites and serve as key regional deposition parameters for input into the PBPK model.

2.3. Pulmonary PBPK model of inhaled rhIFN- α 1b in mice

2.3.1. Systemic and general tissue model structure

The schematic diagram of the model structure is shown in Fig. 2. Each organ, except the lungs, is divided into three compartments: plasma, interstitial space, and endothelial cell compartment. The in vivo transport and disposition of rhIFN- α 1b primarily involve three mechanisms (Branca et al., 1983; Kagan et al., 2010; Kushnaryov et al., 1985; Marijanovic et al., 2007; Sepp et al., 2015; Zanin et al., 2020): (1) Free drug enters the interstitial space through a two-pore model and is subsequently cleared by lymphatic return. The reflection coefficient (σ_v) is used to characterize the resistance of proteins to convective flow, while systemic clearance is simplified to the clearance rate (CL_v); (2) The drug is taken up into the endothelial cell compartment via nonspecific endocytosis and undergoes degradation in the endosome through a first-order rate process; (3) TMDD of rhIFN- α 1b depends on its binding to the type I interferon receptor (IFNAR) and subsequent endocytosis into the lysosome for degradation. Given that IFNAR is widely expressed in pulmonary epithelium and endothelial cells, the receptor interaction region is set in the pulmonary interstitial space to represent drug-receptor interactions after crossing the epithelial barrier (Mahlaköiv et al., 2015; Stanifer et al., 2020). Additionally, due to the rapid binding of rhIFN- α 1b to IFNAR2, which is not the rate-limiting step, the complex receptor complex formation process is simplified in this model to a single binding kinetic process between rhIFN- α 1b and IFNAR1

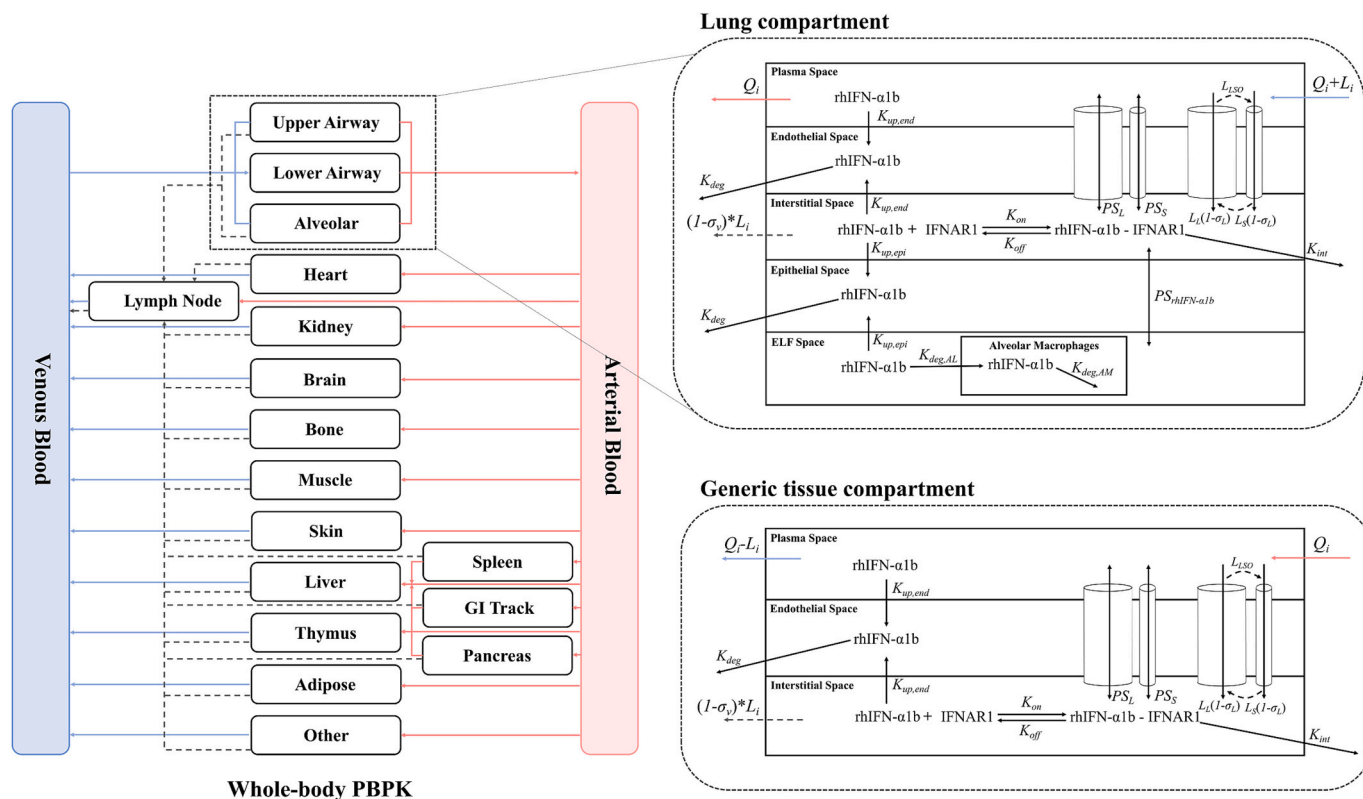


Fig. 2. The structure of pulmonary PBPK model of IFN- α 1b. General tissues are modeled as three compartments, comprising plasma, endothelial endosomes, and interstitial space. The pulmonary tissue is partitioned into three regional compartments: the upper airway, the lower airway, and the alveolar region; each of these is further subdivided into five compartments: plasma, endothelial endosomes, interstitial space, epithelial endosomes, and epithelial lining fluid.

(Kalra et al., 2019; Volkova et al., 2025). The concentration of IFNAR1 in each organ is estimated based on transcriptomic mRNA expression data (Sharma et al., 2025), with liver IFNAR1 abundance (0.242 nM) (Kalra et al., 2025) used as a reference for calculating the IFNAR1 levels in other tissues using the following proportional formula. The mass balance of the drug within each compartment is described by a set of ordinary differential equations, with detailed expressions provided in the supplementary methods 1.3.

$$T_{\text{IFNAR1}} = \frac{n\text{TPM}_T}{n\text{TPM}_L} \times L_{\text{IFNAR1}}$$

Here, T_{IFNAR1} represents the concentration of IFNAR1 in various tissues, measured using a relative transcriptomics approach, while L_{IFNAR1} refers to the reference concentration of IFNAR1 in the liver. Mouse mRNA transcript data, including the normalized expression levels (nTPMT) of IFNAR1 in various tissues and in the liver, were obtained from online transcriptomic databases (<https://www.ebi.ac.uk/gxa/home>).

2.3.2. Lung compartment structure

This study adapted the mouse lung model of 23 airway generations proposed by Lung model for deposition prediction of Balb/c mice in MPPD software (Table S1), dividing the lung into upper airways (generations 1–10), lower airways (generations 11–15), and alveoli (generations 16–23) (Oldham and Robinson, 2007). And this model was originally established based on the morphometric lung model with 23 airway generations reported by Oldham and Robinson (2007), and has been further validated and refined by Yeh et al. and subsequent studies (Oldham and Robinson, 2007; Asgharian et al., 2014; Madl et al., 2010). It enables consistent and comparable predictions of pulmonary deposition in rats, mice and humans. In addition to the plasma, endothelial endosome, and interstitial compartments, each lung compartment was further subdivided into epithelial endosome and epithelial lining fluid (ELF) compartments to describe drug disposition within the epithelial cells and the airway epithelial lining fluid. Free drug enters the lung interstitial space via passive transport mechanisms from ELF. The trans-epithelial transport of free drug between ELF and the interstitial space is described by empirical equations based on molecular weight (Eq. 9 and 10 in supplementary materials). The primary clearance process for free drug in the alveolar epithelial lining fluid is described by Eq. 12 (supplementary materials). Based on the anatomical parameters of the 23-generation mouse airway and using the method proposed by Haini et al., the volume ratios for the upper airways, lower airways, and alveolar compartments were calculated as 1:1.3:8.6. These ratios were used to distribute the volumes of the sub-compartments. Detailed anatomical parameters of the lung are provided in the supplementary method 1.3 (Oldham and Robinson, 2007; Campbell et al., 2015; Fröhlich et al., 2016; MacDonald et al., 1986).

The distribution of pulmonary blood flow across regions is as follows: alveoli 92.5%, upper airways 2.5%, and lower airways 5%. The physiological parameters for the entire mouse model and for the lungs are listed in Tables S2 and S3, while the distribution of IFNAR1 receptors is shown in Table S4, and drug-specific parameters are presented in Table 1. The differential equations describing the transport of rhIFN- α 1b in lung tissue were coded, solved, and simulated using the rxode2 package (version 3.0.4) in R (version 4.3.2). The specific differential equations are provided in the supplementary method 1.3.

2.3.3. Model validation and optimization

This study conducted pharmacokinetic experiments in mice to evaluate the pharmacokinetic characteristics of rhIFN- α 1b in ELF and serum following intratracheal administration, which were used to validate and optimize the mouse pulmonary PBPK model. The experiment utilized female BALB/c mice, aged 8–10 weeks, with a body weight of 20 ± 2 g. rhIFN- α 1b was administered via a customized liquid aerosol delivery device for mice (HRH-MAG4, Beijing Huironghe Technology Co.,

Table 1
Drug-specific parameters of rhIFN- α 1b.

Parameters	Description	Values	Ref
MW (kDa)	Molecular weight	19.3	(Kiem and Schentag, 2008)
$K_{\text{up, end}}$ (1/h)	Endothelial uptake rate	2	Fitted
$K_{\text{up, epi}}$ (1/h)	Epithelial uptake rate	2	Fitted
K_{syn} ($\mu\text{M}/\text{h}$)	IFNAR1 synthesis rate	2.304×10^{-4}	(Branca et al., 1983)
K_{out} (1/h)	IFNAR1 degradation rate	1.38	(Branca et al., 1983; Kalra et al., 2019)
K_{D} (μM)	Affinity constant of rhIFN- α 1b to IFNAR1	0.2	(Kalra et al., 2019)
K_{on} (L/ $\mu\text{mol} \cdot \text{h}$)	Association rate of rhIFN- α 1b to IFNAR1	3.6×10^5	(Kalra et al., 2019)
K_{off} (1/h)	Dissociation rate constant of rhIFN- α 1b from IFNAR1	$K_{\text{off}} = K_{\text{on}} \times K_{\text{D}}$	–
FR_{end}	Endothelial uptake coefficient	0.715	(Djukanović et al., 2014)
FR_{epi}	Epithelial uptake coefficient	0.715	
K_{deg} (1/h)	Clearance of free rhIFN- α 1b in epithelium and endothelium	135.8	
K_{int} (1/h)	Internalization rate of the rhIFN- α 1b-receptor complex from interstitial compartment into epithelium and endothelium	535.86	(Kalra et al., 2019)
a_{e} (nm)	Stokes radius	2.2	(Sepp et al., 2015)
R_{tot} (nM)	Total pulmonary IFNAR1 concentration	0.126–0.155	(Branca et al., 1983; Kalra et al., 2019; Volkova et al., 2025)
$K_{\text{deg, AM}}$ (1/h)	Alveolar macrophage clearance rate	0.001	Fitted
CL_{v} (L/h)	Systemic clearance	Mouse: 0.003 Adult: 1.08	Fitted and translated
$K_{\text{abs, UA}}$ (1/h)	Paracellular transport rate between ELF and interstitial compartment in the upper airway	0.096	Eq.9 (Djukanović et al., 2014)
$K_{\text{abs, LA}}$ (1/h)	Paracellular transport rate between ELF and interstitial compartment in the lower airway		
$K_{\text{abs, AL}}$ (1/h)	Paracellular transport rate between ELF and interstitial compartment in the alveolar region		
$F_{\text{lu, drug}}$	Fraction of drug transferred from ELF to interstitial compartment	Eq.11	
$K_{\text{deg, AL}}$ (1/h)	First-order degradation rate constant of rhIFN-	Eq.12	

(continued on next page)

Table 1 (continued)

Parameters	Description	Values	Ref
$PS_{drug, UA}$ (L/h)	$\alpha 1b$ in the ELF of the alveolar region	Eq.10	
	Permeability-surface area		
	product of the drug between ELF and interstitial compartment in the upper airway		
$PS_{drug, LA}$ (L/h)	Permeability-surface area	Eq.10	
	product of the drug between ELF and interstitial compartment in the lower airway		
	Permeability-surface area		
$PS_{drug, AL}$ (L/h)	product of the drug between ELF and interstitial compartment in the alveolar region	0.2	
	Lymphatic reflection coefficient		

China) through a single intratracheal dose. The dose was normalized based on the adult clinical dose using the body surface area method, with a target dose of 23.7 $\mu\text{g/kg}$ and a drug concentration of 15 $\mu\text{g/mL}$. Serum and bronchoalveolar lavage fluid (BALF) samples were collected at time points between 0.25 and 12 h post-administration, with 4 mice per time point, totaling 28 mice. rhIFN- $\alpha 1b$ concentrations in serum and BALF were quantified using a validated double-antibody sandwich ELISA method. Urea concentrations in serum and BALF were measured using a urea nitrogen assay kit (BC1535-100 T/48S, Beijing Solarbio Technology Co., China), which were used to calculate the corrected drug concentration in ELF. This study was approved by the Animal Welfare and Ethics Committee of Central South University (APU-2026-0046), and the experimental procedures strictly adhered to animal ethical guidelines and operational protocols. Detailed experimental methods are provided in the supplementary method 1.4.

The deposition data input into the mouse pulmonary PBPK model were simulated using the MPPD software. The BALB/c mouse airway model was selected, with upright posture and intratracheal administration. However, the exact MMAD under the specific experimental condition could not be directly measured due to experimental limitations. Therefore, the MMAD and GSD parameters were taken from the experimental values of the 15 $\mu\text{g/mL}$ group in the in vitro aerosol study as surrogate input parameters. Default respiratory parameters from the software were used, and the model calculated the deposition fractions in the upper airways, lower airways, and alveolar region, which were then input into the PBPK model. The model was optimized using mouse pharmacokinetic data, with CL_v estimated based on serum observations. After fitting the serum data, the key parameters for lung disposition, such as the nonspecific epithelial uptake rate ($K_{up, epi}$) and nonspecific endothelial uptake rate ($K_{up, end}$), were further estimated using ELF observations.

2.3.4. Parameter sensitivity analysis

A local sensitivity analysis was conducted in this study to evaluate the impact of various parameters on serum and pulmonary tissue exposure. Given the high interrelationship of parameters in the drug transport process, we selected the following parameters for analysis: alveolar ELF free drug clearance rate ($K_{deg, AL}$), $K_{up, epi}$, $K_{up, end}$, drug-receptor affinity (K_{on}), upper and lower airway epithelial paracellular

transport rates ($K_{abs, UA}$, $K_{abs, LA}$), endosomal clearance rates in epithelial and endothelial cells (K_{deg}), and CL_v . By varying each parameter by $\pm 50\%$ from its baseline value, we calculated the percentage change in AUC_{0-inf} for both serum and ELF, thereby quantifying the sensitivity of the model output to changes in each parameter.

2.4. Translation of the adult pulmonary model for inhaled rhIFN- $\alpha 1b$

2.4.1. Model structure and parameters

After constructing and validating the mouse pulmonary PBPK model, the fitted $K_{up, epi}$ and $K_{up, end}$ values were fixed, and CL_v was extrapolated to the adult model using allometric scaling methods (Zou et al., 2012). The mouse model's physiological parameters were then replaced with the corresponding adult parameters, thereby translating the mouse model into an adult pulmonary PBPK model. The compartment structure of the adult model remained consistent with the validated mouse model, with the addition of a macrophage compartment in the alveolar ELF compartment to account for macrophage-mediated clearance, with a macrophage volume set at 0.549 mL (Kiem and Schentag, 2008). Based on the Weibel model, the human airway structure was divided into 0–23 generations. Anatomically and functionally, generations 0–9 were designated as the upper airways, generations 10–16 as the lower airways, and generations 17–23 as the alveoli. The systemic physiological parameters were integrated from studies by Dhaval et al., and were further adjusted according to authoritative physiological databases, including the International Commission on Radiological Protection (ICRP) Publication 89 (Shah and Betts, 2012; Davies and Morris, 1993). Pulmonary regional physiological parameters were calculated using anatomical data from the Weibel model, employing the method proposed by Haini et al. The volume ratio for the upper airways, lower airways, and alveoli was set at 1:2.2:7.8, with corresponding sub-compartment volumes allocated accordingly. The calculated values were detailed in the supplementary method 1.5 (Campbell et al., 2015; Fröhlich et al., 2016; MacDonald et al., 1986; Ayier et al., 1986). The physiological parameters for the entire body and sub-compartments of the lung model are summarized in Tables S5 and S6, with the relevant drug-specific parameters remaining consistent, as shown in Table 1. The anatomical data for the standard adult Weibel model are provided in Table S7, and the calculated concentrations of IFNAR1 in the interstitial compartments of various organs are presented in Table S8.

2.4.2. Model validation

We conducted a pharmacokinetic study of rhIFN- $\alpha 1b$ inhalation solution in healthy adults (<https://www.chictr.org.cn/>, CRT20231702). Two dose groups were set at 12×10^5 IU, 18×10^5 IU, with a single inhalation via facemask. The study collected concentration data for rhIFN- $\alpha 1b$ in serum, ELF, and AM of healthy adults. Additionally, we gathered serum data for rhIFN- $\alpha 1b$ from the literature following a single nebulized inhalation in healthy adults. The dosing regimen involved single inhalations via mouthpiece at doses of 2×10^5 IU, 6×10^5 IU, 12×10^5 IU, 18×10^5 IU (Peng et al., 2024). The trial designs for both studies are summarized in Table 2.

By inputting the dosing regimens and deposition data, the pulmonary PBPK model of rhIFN- $\alpha 1b$ in healthy adults simulated the concentration-time profiles in serum, ELF, and AM. The model predictions were compared with the observed values. If the ratio of predicted to observed values for key pharmacokinetic parameters, C_{max} and AUC_{0-inf} , fell within the validation range of 0.5 to 2.0, the model's predictive ability was considered satisfactory.

3. Results

3.1. Aerosol characterization of nebulized rhIFN- $\alpha 1b$

The particle size distribution of nebulized rhIFN- $\alpha 1b$ (Table S9) showed no significant differences in D50 among concentration groups

Table 2
Study design of two clinical studies for model validation.

No.	Sample size	Male, %	Dosing Regimen	Sampling points	Age, yrs	Weight, kg	Ref
1	40	100%	12×10^5 IU, 18×10^5 IU, inhalation 15–20 min (mask), single dose	Serum: 0.5, 1, 1.5, 2, 3, 4, 6, 8, 12, 24, 48 h after end of inhalation; BALF/AM: 1, 2, 3, 4, 6, and 12 h after the end of administration.	26.5 (18–40)	65.9 ± 6.72	CRT20231702
2	16	68.75%	2×10^5 IU, 6×10^5 IU, 12×10^5 IU, 18×10^5 IU, inhalation 10–15 min (mouthpiece), single dose*	Serum: 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4, 6, 8, 12, 24, 48 h after end of inhalation	28.6 (20–42)	61.6 ± 6.87	(Kiem and Schentag, 2008)

BALF: Bronchoalveolar Lavage Fluid; AM: Alveolar Macrophage.

* Due to the unverified dosing frequency of the multiple ascending dose arm in the Peng et al. (2024) study, the analysis was restricted to single ascending dose data.

($P > 0.05$), although the 30 $\mu\text{g/mL}$ group exhibited the smallest D50, at 2.65 μm (Fig. S1a). For dQ3 (1–5 μm), significant differences were observed among selected groups ($P < 0.05$), with the highest value at 30 $\mu\text{g/mL}$ ($54.62 \pm 0.16\%$, Fig. S1b). As shown in Fig. 3a, the aerosol exhibited a bimodal volume distribution, with peaks below 1 μm and at 4–6 μm .

Particles were primarily deposited in the 2nd–5th stages of the NGI (Fig. 3b, c). Two-way ANOVA indicated no significant effects of inhalation device, drug concentration, or their interaction on aerodynamic parameters ($P > 0.05$, Table S10). Accordingly, MMAD, GSD, and FPF remained stable across all groups (5.16–5.68 μm , 1.85–2.15, and 41.79%–47.16%, respectively), despite a slight tendency toward smaller MMAD/GSD and higher FPF at 15–30 $\mu\text{g/mL}$ (Fig. S1c–e).

Two-way ANOVA showed that both inhalation device and drug concentration significantly affected aerosol delivery rate, total delivery, and delivery efficiency ($P < 0.005$, $P < 0.0001$), with no interaction between factors (Table S11). At the same concentration, the mouthpiece consistently outperformed the facemask. Device-related differences in delivery rate were significant only at 60 $\mu\text{g/mL}$ (Fig. S2a), whereas differences in total delivery and efficiency were observed at 45 $\mu\text{g/mL}$ (Fig. 3d, S2b). With increasing concentration, delivery rate and total delivery increased nonlinearly (Fig. S2c–d), while delivery efficiency decreased (Fig. 3e–f). Overall delivery efficiency ranged from 11.06% to 21.66%.

3.2. Pulmonary deposition prediction of nebulized rhIFN- $\alpha 1b$

Based on NGI experiments and MPPD model predictions, regional deposition of rhIFN- $\alpha 1b$ differed markedly between inhalation devices. The mouthpiece significantly increased deposition in the lower airways and alveolar regions compared with the facemask (Table 3). With facemask inhalation, aerosol deposition was predominantly retained in the oropharynx (52.77%–55.26%), with limited deposition in the upper airways (6.34%–6.73%), lower airways (5.15%–5.74%), and alveoli (13.75%–14.75%). In contrast, the mouthpiece reduced oropharyngeal deposition to 23.70%–27.19% and increased alveolar deposition to 19.90%–22.31%, approximately 1.5-fold higher than that of the facemask. These differences indicate improved peripheral lung delivery with the mouthpiece.

3.3. Pulmonary PBPK model of inhaled rhIFN- $\alpha 1b$ in mice

CL_v was estimated from mouse serum pharmacokinetic data as 3 mL/h. The transmembrane transport parameters ($K_{up, \text{epi}}$ and $K_{up, \text{end}}$) were assumed to be equal and estimated as 2 h^{-1} based on ELF data. After parameter estimation, the model fit both serum and ELF rhIFN- $\alpha 1b$ concentration data well (Fig. S3). The C_{max} and $AUC_{0-\text{inf}}$ ratios of predicted to observed values were within the acceptable range of 0.5–2 (Table S14).

Sensitivity analysis showed that systemic exposure was most

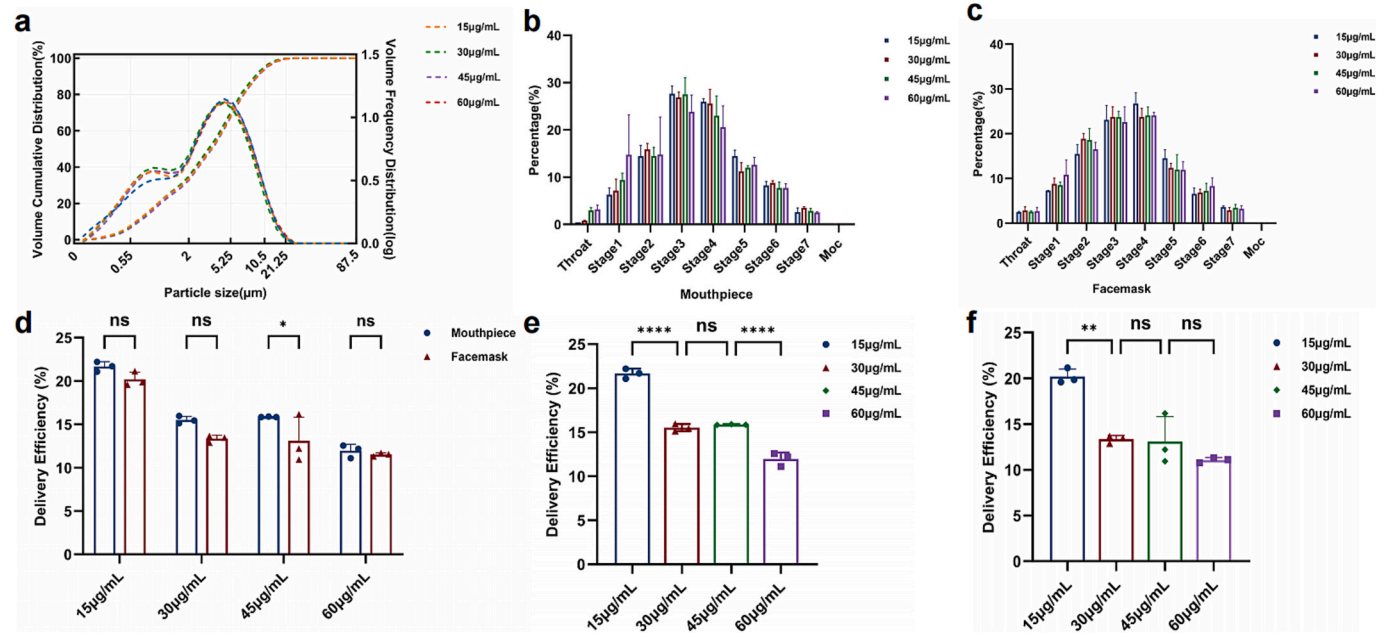


Fig. 3. Evaluation of the aerosol delivery performance of IFN- $\alpha 1b$. (a) Cumulative particle size distribution curves and frequency distributions at various concentrations measured by laser diffraction. Proportion of drug deposited on each stage of the NGI using a mouthpiece (b) and a facemask (c). (d) Effect of different inhalation devices on delivery efficiency. Effect of drug concentration on delivery efficiency when using a mouthpiece (e) and a facemask (f).

Table 3
MPPD-predicted drug deposition.

Inhalation device	Concentration ($\mu\text{g/mL}$)	MMAD (μm)	GSD	Oropharynx	Upper airway	Lower airway	Alveolar region
Mouthpiece	15	5.27 ± 0.23	1.85 ± 0.09	23.70%	11.17%	9.98%	22.31%
	30	5.43 ± 0.32	1.90 ± 0.08	25.17%	11.56%	9.57%	21.34%
	45	5.63 ± 0.47	1.97 ± 0.11	26.99%	11.95%	8.94%	20.24%
	60	5.61 ± 0.80	2.01 ± 0.16	27.19%	11.94%	8.73%	19.90%
Facemask	15	5.16 ± 0.34	2.02 ± 0.10	52.77%	6.34%	5.74%	14.75%
	30	5.68 ± 0.09	1.99 ± 0.08	55.26%	6.73%	5.51%	14.12%
	45	5.59 ± 0.38	2.01 ± 0.11	54.75%	6.68%	5.51%	14.18%
	60	5.53 ± 0.42	2.15 ± 0.20	53.53%	6.69%	5.15%	13.75%

MMAD: Mass Median Aerodynamic Diameter; GSD: Geometric Standard Deviation.

sensitive to CL_v , followed by $K_{abs, UA}$, whereas $K_{up, epi}$ and $K_{up, end}$ had relatively smaller effects. In contrast, ELF exposure was primarily sensitive to $K_{deg, AL}$ and $K_{up, epi}$, with minimal influence from $K_{up, end}$.

3.4. Pulmonary PBPK model of inhaled rhIFN- $\alpha 1b$ in healthy adults

PBPK model inputs incorporated region-specific deposition fractions derived from delivery efficiency and MPPD simulations of Inhaled rhIFN- $\alpha 1b$ (Table S13). Compared with the facemask, the mouthpiece showed higher overall delivery efficiency (21.66% vs 9.57%) and increased deposition in the lower airways and alveolar region (3.11% and 9.31% vs 0.69% and 2.29%, respectively), while oropharyngeal deposition remained comparable (~6%).

CL_v estimated from mouse data was extrapolated to humans using allometric scaling, yielding an adult value of 1.08 L/h, while $K_{up, epi}$ and $K_{up, end}$ values remained unchanged. Replacement of mouse physiological parameters with those of healthy adults enabled species translation from mice to humans. The adult model was able to accurately predict the serum concentrations of rhIFN- $\alpha 1b$ after a single inhalation of 2×10^5 , 6×10^5 , 12×10^5 , 18×10^5 IU rhIFN- $\alpha 1b$ using a mouthpiece (Fig. 4a–d), as well as the serum and ELF drug concentrations after mask inhalation of 12×10^5 and 18×10^5 IU (Fig. 4e, g). Given that AM uptake is a key mechanism for the clearance of therapeutic proteins in the alveolar region, the adult PBPK model incorporated an AM compartment and quantitatively characterized the AM-mediated uptake and clearance rate of rhIFN- $\alpha 1b$ based on clinical observational data. The model successfully simulated the AM drug concentration changes at 12×10^5 and 18×10^5 IU doses (Fig. 4f, h). Diagnostic plots for model fitting accuracy and normalized residual analysis (Figs. S7 and S8) are included in the

Supplementary Material.

The model predicted the C_{max} and AUC_{0-inf} of rhIFN- $\alpha 1b$ in serum, ELF, and AM with ratios of predicted to observed values generally within the 0.5–2.0 range (Table 4). While the AUC_{0-inf} ratio of 2×10^5 IU mouthpiece group was 2.63, which may be attributed to the small sample size ($n = 2$) for the observed data. For the mask inhalation of 12×10^5 and 18×10^5 IU, the C_{max} ratios for AM were 0.49 and 0.4, respectively, while AUC_{0-inf} ratios remained within the 0.5–2.0 range. Overall, the model adequately described the pharmacokinetics of rhIFN- $\alpha 1b$ across inhalation devices and dose levels in both systemic circulation and pulmonary compartments.

4. Discussion

Despite the rapid advancement in inhaled therapeutic proteins, pulmonary pharmacokinetic studies of these agents remain limited, largely due to the lack of quantitative models capable of integrating in vitro delivery with in vivo disposition (Onoue et al., 2026). Conventional approaches are not sufficient to simultaneously characterize the delivered dose, pulmonary deposition, and subsequent systemic transfer, which limits a comprehensive understanding of the in vivo behavior of inhaled therapeutic proteins. To address this gap, we developed a pulmonary PBPK model for rhIFN- $\alpha 1b$ that incorporates in vitro aerosol deposition data and in vivo pharmacokinetic profiles, and enables extrapolation across species from mice to healthy adults. The model was evaluated using both preclinical and clinical data and showed good predictive performance for drug exposure in the lung and systemic circulation.

To ensure quantitative precision in the mouse pulmonary PBPK

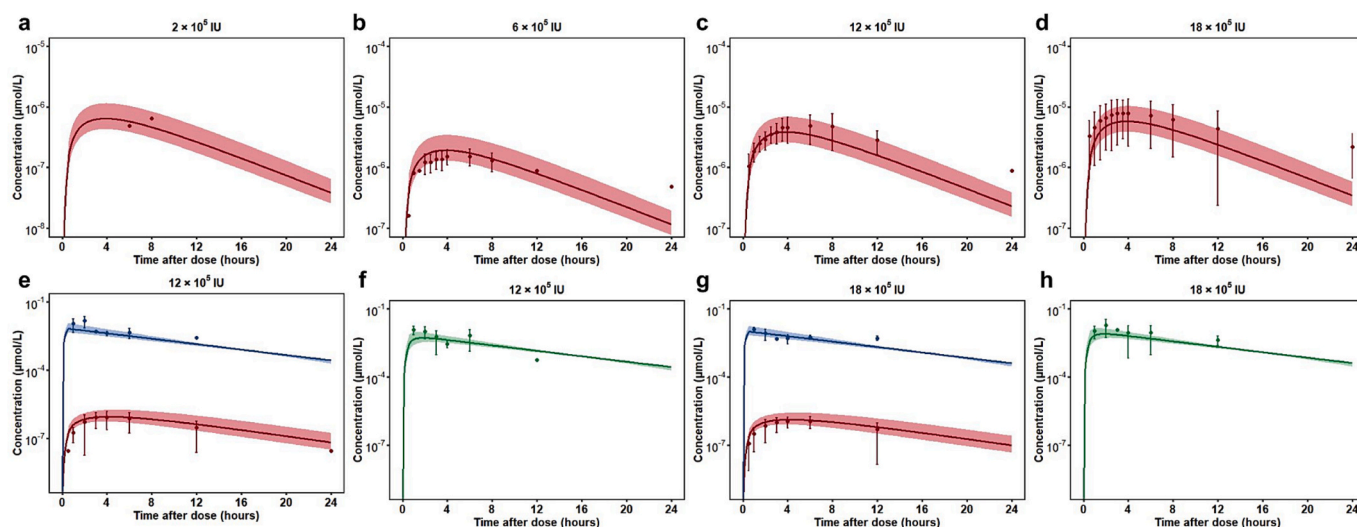


Fig. 4. Predicted pharmacokinetic profiles of IFN- $\alpha 1b$ using the adult pulmonary PBPK model. (a–d) Administration via a mouthpiece. (e–h) Administration via a facemask. Circles represent observed values (red: serum; blue: ELF; green: AM). Solid lines indicate the mean model-predicted curves, and shaded areas represent the 5%–95% prediction intervals. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4

Observed-to-predicted ratios of C_{max} and AUC_{0-inf} for the adult pulmonary PBPK model of rhIFN α 1b.

Study	Dose, IU	Matrix	Parameters, units	Observed	Predicted	Ratio	
Peng et al. 2024 (Kiem and Schentag, 2008)	2*10 ⁵	Serum	C _{max} , pg /mL	11.3	11.78	1.04	
			AUC _{0-inf} , h*pg/mL	21.5	56.59	2.63	
	6*10 ⁵	Serum	C _{max} , pg /mL	30.513	44.19	1.45	
			AUC _{0-inf} , h*pg/mL	293.613	511.63	1.74	
	12*10 ⁵	Serum	C _{max} , pg /mL	104.9	88.18	0.84	
			AUC _{0-inf} , h*pg/mL	1295.35	1021.32	0.79	
	18*10 ⁵	Serum	C _{max} , pg /mL	172.375	132.36	0.77	
			AUC _{0-inf} , h*pg/mL	1897.95	1532.95	0.81	
	CRT20231702		Serum	C _{max} , pg /mL	21.09	18.64	0.88
				AUC _{0-inf} , h*pg/mL	149.436	215.12	1.44
12*10 ⁵		ELF	C _{max} , pg /mL	294,733.5	135,658.91	0.46	
			AUC _{0-inf} , h*pg/mL	1,239,253	988,372.09	0.80	
		AM	C _{max} , pg /mL	237,612	116,279.07	0.49	
			AUC _{0-inf} , h*pg/mL	1,032,009	930,232.56	0.90	
		Serum	C _{max} , pg /mL	25.43	27.91	1.10	
			AUC _{0-inf} , h*pg/mL	169.625	323.64	1.91	
18*10 ⁵		ELF	C _{max} , pg /mL	256,193.8	213,178.29	0.83	
			AUC _{0-inf} , h*pg/mL	1,427,806	1,472,868.22	1.03	
	AM	C _{max} , pg /mL	383,658.5	155,038.76	0.40		
		AUC _{0-inf} , h*pg/mL	2,060,625	1,375,968.99	0.67		

ELF: Epithelial Lining Fluid; AM: Alveolar Macrophage; C_{max} : Peak Concentration; AUC_{0-inf} : Area Under the Curve from time zero to infinity.

model, we addressed parameter uncertainty regarding the HRH-MAG4 nebulizer's aerosol characteristics by employing MMAD values derived from in vitro characterization of the same drug concentration. Subsequent sensitivity analysis using MPPD simulations (MMAD 5–7 μ m, GSD 1.85) demonstrated that the deposition profile remained remarkably stable within this range, with over 87% of the dose consistently depositing in the upper airways (Fig. S9). This insensitivity is attributed to the predominant role of inertial impaction during intratracheal administration. In the restricted anatomical space of the mouse trachea (~1 mm), the high-velocity discharge from the nebulizing needle causes the Stokes number to surge (Anjilvel and Asgharian, 1995; Bivas-Benita et al., 2005; Kishimoto and Morimoto, 2021). Consequently, micron-sized droplets are intercepted by the proximal airway walls before they can reach the distal lung tissue. This device-dependent deposition pattern masks the subtle regulatory effects of particle size, thereby ensuring the robustness of our model predictions against MMAD fluctuations. Due to the high stability of such deposition profiles, the PBPK model incorporating these simulated deposition parameters yielded plasma concentration–time profiles with high fidelity to the observed PK data (Fig. S10).

In the validation of the pulmonary PBPK model in adults, it should be noted that the AUC_{0-inf} ratio of 2×10^5 IU mouthpiece group reached 2.63, exceeding the commonly used two-fold acceptance criterion. This discrepancy is most likely attributable to the very small sample size ($n = 2$) in this group. With such limited samples, inter-individual variability could not be adequately characterized, which reduced the representativeness of the observed mean value and consequently led to a larger deviation from the model-predicted AUC. Importantly, the observed concentrations still fell within the 5th–95th percentile prediction interval of the population simulation. Therefore, when considering the overall model performance across all datasets and the satisfactory coverage of the prediction intervals, the present PBPK model is still considered to have reasonable predictive capability for rhIFN- α 1b.

Notably, clear differences were observed between mask and mouthpiece inhalation devices in terms of both pulmonary bioavailability and pharmacokinetics. Using the PBPK model, these differences could be quantitatively described and further interpreted mechanistically. Compared with mask administration, mouthpiece delivery resulted in higher pulmonary deposition. This is largely due to the mixed oral-nasal breathing pattern during mask use, which leads to substantial drug loss in the nasal cavity and oropharynx. For macromolecules such as therapeutic proteins, absorption across the nasal mucosa is extremely limited. Therefore, this portion of the deposition contributes

little to systemic exposure (Ozsoy et al., 2009; Ugwoke et al., 2001). In contrast, mouthpiece delivery reduces upper airway losses and increases the fraction of drug reaching the lungs, thereby improving systemic exposure. Beyond delivery efficiency, the distribution of drug within the lung also matters. Regional differences in blood flow, epithelial permeability, and cellular composition lead to heterogeneous absorption across the lung (Patton and Byron, 2007). As a result, variations in deposition patterns introduced by different delivery devices may translate into differences in the pharmacokinetic profile.

PBPK modeling has the potential to facilitate the development of locally administered drugs. However, pulmonary PBPK models for inhaled therapeutic proteins are still lacking. Although Sharad Sharma et al. established a minimal pulmonary PBPK model in mice using intravenous antibody data, the lung was simplified as a single compartment with an assumed homogeneous distribution, limiting its ability to capture regional differences in deposition and disposition (Sharma et al., 2023). Similarly, Dhaval Patel et al. developed a compartmentalized human lung PBPK model, but the model was mainly validated with intravenous monoclonal antibody data, and its applicability to inhaled delivery remains to be fully demonstrated (Jagdale et al., 2022). In this study, we developed a pulmonary PBPK model specifically for inhaled therapeutic proteins. The model links aerosol deposition with pulmonary disposition processes, allowing quantitative description of drug kinetics in both the lung and systemic circulation, and supporting extrapolation from preclinical species to humans.

Although this study successfully established a quantitative translational framework for inhaled therapeutic proteins, there remain areas for further refinement in future research. Although the use of the gold-standard NGI device and conventional connection tubing ensured high reproducibility, incorporating biomimetic oropharyngeal models would better replicate human anatomy and enhance the physiological relevance of in vitro deposition data. Additionally, although the current PBPK model was validated using rhIFN- α 1b as a representative therapeutic protein and demonstrated satisfactory predictive performance, different therapeutic proteins may possess distinct mechanisms of action. For example, monoclonal antibodies, which have been extensively studied, undergo FcRn-mediated recycling that significantly influences their pharmacokinetics. Due to the current scarcity of validation data for other protein drugs, these specific mechanisms have not yet been incorporated into the model. In future studies, once sufficient validation data become available, these mechanisms can be integrated and validated to further expand the applicability of the model.

5. Conclusion

In summary, this study is the first to construct and validate an inhaled therapeutic protein pulmonary PBPK model that integrates in vitro aerosol characterization results with interspecies extrapolation methods. The model effectively describes the pulmonary and systemic disposition of inhaled therapeutic proteins. The established model framework successfully integrates in vitro deposition characteristics with in vivo pharmacokinetic behaviors, providing important insights for the clinical translation of inhaled therapeutic proteins, initial dose selection, and optimization of dosing regimens.

Author contributions

Jin-long Liu and Ya-xin Liu are responsible for conducting experiments, data analysis, model development, and writing of the manuscript; Guang-lin Wang, Han-feng Zhang, Nuo Xu, Sheng-nan Zhang, Xiao Yue, Xu-xia Liang and Yin-jia Luo are responsible for assisting in conducting experiments and data analysis; Xue-juan Zhang, Heng-xin Peng and Qi Pei are responsible for guidance and review of the experimental protocol; Yun Kuang and Guo-ping Yang are responsible for research design, guidance and review of the experimental protocol.

CRediT authorship contribution statement

Jin-long Liu: Writing – original draft, Project administration, Investigation, Data curation. **Ya-xin Liu:** Writing – review & editing, Writing – original draft, Project administration, Formal analysis, Data curation. **Xue-juan Zhang:** Project administration. **Guang-lin Wang:** Project administration. **Jie Huang:** Project administration. **Han-feng Zhang:** Project administration. **Nuo Xu:** Project administration. **Sheng-nan Zhang:** Project administration. **Xiao Yue:** Project administration. **Xu-xia Liang:** Project administration. **Yin-jia Luo:** Project administration. **Heng-xin Peng:** Resources. **Qi Pei:** Writing – review & editing. **Yun Kuang:** Writing – review & editing, Funding acquisition, Conceptualization. **Guo-ping Yang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Peng Hengxin is an employee of Kexing Biopharm Co., Ltd. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijpx.2026.100586>.

Data availability

The datasets used and/or analyzed during the current study are

available from the corresponding author on reasonable request.

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